HR ANALYTICS PROJECT

Problem Statement & Project Scope Document

1. Background & Industry Context

Employee attrition is one of the most pressing and costly challenges facing organizations across industries today. When talented employees leave, companies bear the direct costs of recruiting, onboarding, and training replacements, as well as indirect costs such as lost productivity, reduced team morale, and knowledge drain. Industry research consistently estimates that replacing a single employee can cost between 50% to 200% of their annual salary, depending on the role and seniority level.

For a mid-to-large enterprise with thousands of employees, even a modest attrition rate can translate into tens of millions of dollars in annual losses. Despite this, many HR departments continue to rely on reactive, intuition-driven decisions rather than proactive, data-backed strategies. The shift toward people analytics and data-driven HR management has become a strategic imperative, not an option.

This project addresses this gap by leveraging the HR Analytics Case Study dataset (vjchoudhary7, Kaggle), which contains real-world-style employee records across three linked tables encompassing 4,410 employees, 30+ combined attributes, and five analytical dimensions: general demographics, job characteristics, compensation, employee survey responses, and manager evaluations.

2. Problem Statement

A large multinational corporation is experiencing an annual employee attrition rate that exceeds the industry benchmark, resulting in significant financial loss, operational disruption, and declining workforce morale. The HR department lacks the analytical capability to identify why employees are leaving, which employee segments are at highest risk, and what organizational or personal factors are the strongest predictors of attrition. Without this intelligence, the company cannot design targeted retention strategies, optimize compensation structures, or build a healthy organizational culture

3. Project Objectives

1. Perform a complete Descriptive Analysis to understand the workforce composition, attrition distribution, and key HR metrics across the organization.
2. Conduct a Diagnostic Analysis to identify correlations, patterns, and root causes that explain why attrition occurs in specific employee segments.
3. Build a Logistic Regression Model to predict the probability of employee attrition and identify the most influential predictor variables.
4. Develop a Power BI Dashboard that enables HR leaders to interactively explore attrition trends, risk segments, and KPIs at a glance.

4. Key Research Questions

This project will answer the following analytical questions:

#	Research Question	Analysis Type
1	What is the overall attrition rate, and how does it vary by department and job role?	Descriptive
2	Which age group, gender, and marital status categories have the highest attrition?	Descriptive
3	How does monthly income and salary hike percentage relate to attrition?	Descriptive / Diagnostic
4	Does working overtime significantly increase the likelihood of an employee leaving?	Diagnostic
5	What is the relationship between job satisfaction, environment satisfaction, and attrition?	Diagnostic
6	Does distance from home correlate with higher attrition rates?	Diagnostic
7	Which employee attributes are the strongest predictors of attrition?	Predictive (Logistic Reg.)
8	Can we accurately predict whether a given employee will leave?	Predictive (Logistic Reg.)
9	What factors predict an employee's monthly income most accurately?	Predictive (Linear Reg.)
10	Which employee segments should HR prioritize for retention interventions?	Dashboard / Insight